Unit: 6 Unit 4: Absolute Value
Topic: Definition of Absolute Value

Name: ________________
Date: ____________

Advanced (Habanero)

Q1. What is the absolute value of -5?
(A) -5
(B) 5
(C) 0
(D) 10
(E) 25

Q2. On a number line, the distance from 0 to -3 is
(A) -3
(B) 3
(C) 0
(D) 6
(E) 9

Q3. Evaluate $|2-5|$
(A) -3
(B) 3
(C) 0
(D) 2
(E) 5

Q4. Solve $|x| = 4$
(A) $x = 4$
(B) $x = -4$
(C) $x = 0$
(D) $x = pm4$
(E) $x = pm2$

Q5. The absolute value of x is $|x| = 7$. Then x could be
(A) 7
(B) -7
(C) 0
(D) 7 or -7
(E) 14

Q6. Which of the following is equivalent to $|3x|$?
(A) $3|x|$
(B) $|3||x|$
(C) $3|x| + 2$
(D) $|3x + 1|$
(E) $3|x| - 2$

Q7. If $|x + 2| = 5$, then x could be
(A) 3
(B) -7
(C) 0
(D) 3 or -7
(E) 1 or -9

Q8. The absolute value of the sum of two numbers is $|a+b| = 10$. Then
(A) $a + b = 10$
(B) $a + b = -10$
(C) $a + b = 0$
(D) $a + b = 10$ or $a + b = -10$
(E) $a = b = 5$

Open Ended Questions (FRQ)

Q9. Evaluate the expression $|2x - 3|$ when $x = -2$

Q10. Solve the equation $|x - 2| = |x + 3|$ and find the value of x

Chef's Tips

- Make sure to check the units of the answers
- Remind students to simplify their expressions
- Encourage students to use number lines to visualize the problems